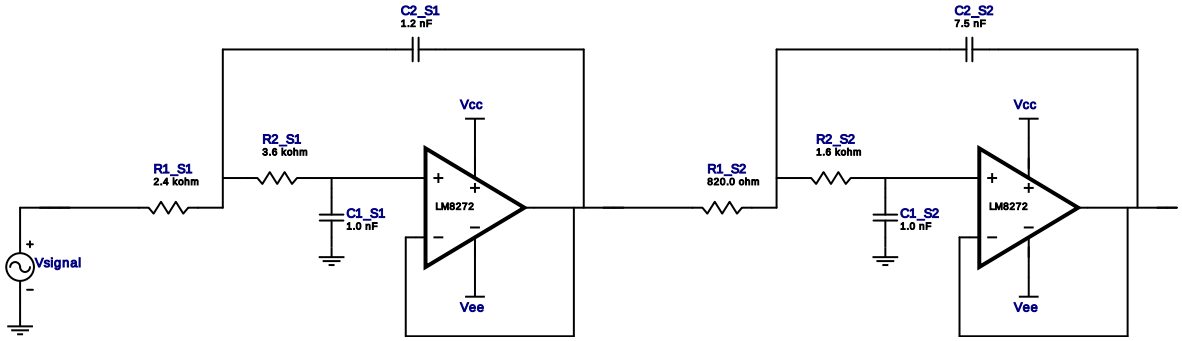
Type : Lowpass
Response : Butterworth
Order : 4
Number of Stages : 2

Filter Design Report

Design : Lowpass Filter - 4th order Butterworth
Design ID: 12



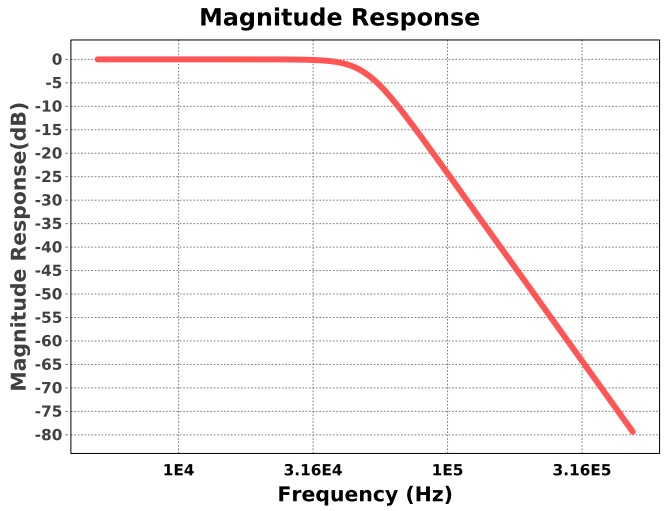
Electrical BOM

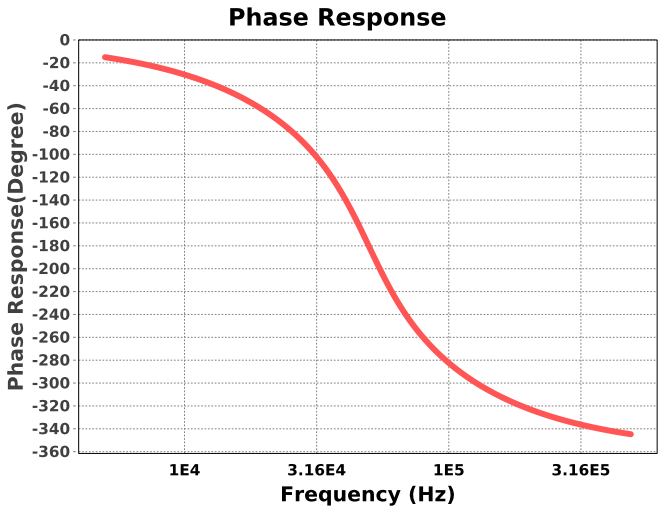
#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	LM8272	GbwTyp= 13MHz VccMax= 24V VccMin= 2.5V	1
2.	A1_S2	Texas Instruments Inc.	LM8272	GbwTyp= 13MHz VccMax= 24V VccMin= 2.5V	1
3.	C1_S1	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
4.	C1_S2	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
5.	C2_S1	Generic	Ideal	Cap= 1.2 nF Tolerance= 2.0 %	1
6.	C2_S2	Generic	Ideal	Cap= 7.5 nF Tolerance= 2.0 %	1
7.	R1_S1	Generic	Ideal	Res= 2400.0ohm Tolerance= 1%	1
8.	R1_S2	Generic	Ideal	Res= 820.0ohm Tolerance= 1%	1
9.	R2_S1	Generic	Ideal	Res= 3600.0ohm Tolerance= 1%	1
10.	R2_S2	Generic	Ideal	Res= 1600.0ohm Tolerance= 1%	1

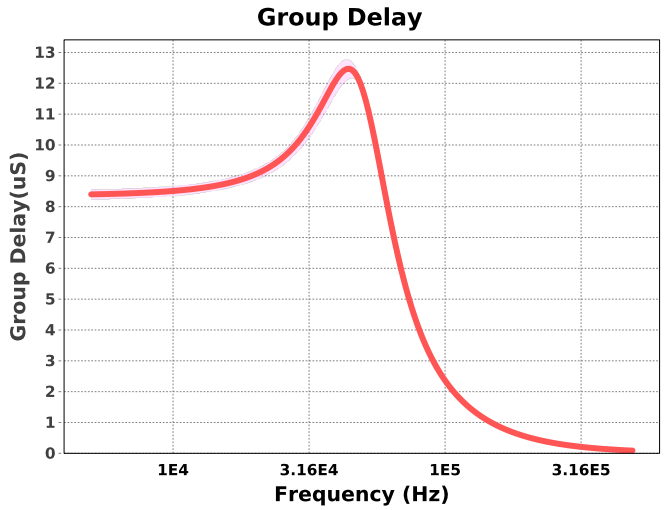
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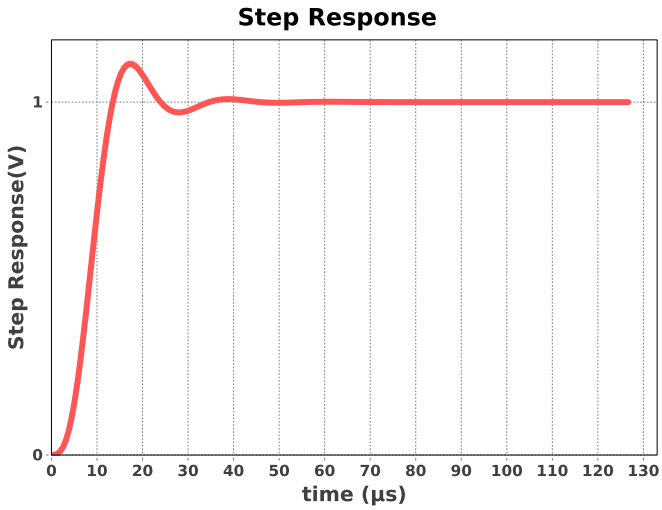
Sensitivity Analysis

#	Name	Series	Tolerance
1.	Cap	E48	2%
2.	Res	E96	1%









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Design Inputs

#	Name	Value	Description
1.	FilterType	lowpass	
2.	FilterResponse	Butterworth	
3.	FilterOrder	4.0	
4.	FilterTopology	Sallen-Key	
5.	NumberOfStages	2.0	
6.	PassbandFrequency	50.0 k	
7.	StopbandAttenuation	-55.917	
8.	StopbandFrequency	250.0 k	
9.	Gain	1.0	
10.	DualSupply	+/-5.00 V	Power supply(s) to active chips
11.	ResistorTolerance	E96	Resistor series - 1% Passive resistor tolerance
12.	CapacitorTolerance	E48	Capacitor series - 2% Passive capacitor tolerance

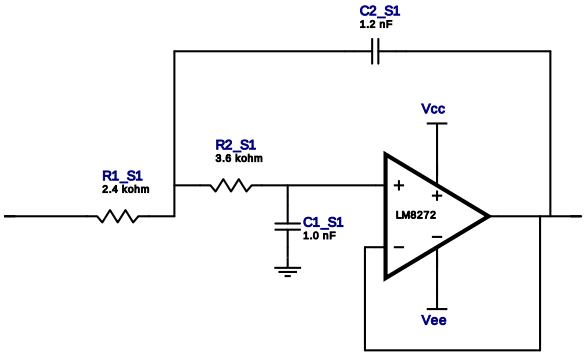
Design Assistance

1. **LM8272** Product Folder : http://www.ti.com/product/LM8272 : contains the data sheet and other resources.

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Filter Stage :1

Cutoff Frequency	49.428 kHz
Min GBW Reqd	2.706 MHz
Stage Gain	1.0 V/V
Stage Q	536.656 m
Stage Topology	Sallen-Key



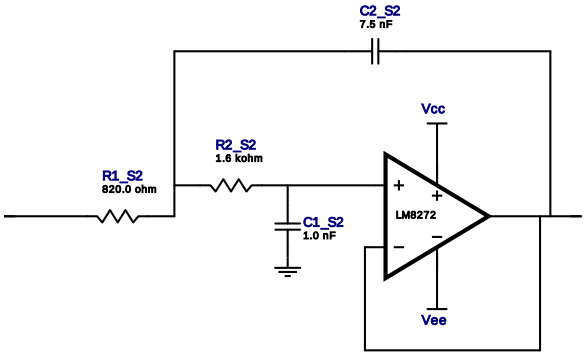
Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	LM8272	GbwTyp= 13MHz VccMax= 24V VccMin= 2.5V	1
2.	C1_S1	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
3.	C2_S1	Generic	Ideal	Cap= 1.2 nF Tolerance= 2.0 %	1
4.	R1_S1	Generic	Ideal	Res= 2400.0ohm Tolerance= 1%	1
5.	R2_S1	Generic	Ideal	Res= 3600.0ohm Tolerance= 1%	1

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Filter Stage :2

Cutoff Frequency	50.737 kHz
Min GBW Reqd	6.533 MHz
Stage Gain	1.0 V/V
Stage Q	1.296
Stage Topology	Sallen-Key



Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S2	Texas Instruments Inc.	LM8272	GbwTyp= 13MHz VccMax= 24V VccMin= 2.5V	1
2.	C1_S2	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
3.	C2_S2	Generic	Ideal	Cap= 7.5 nF Tolerance= 2.0 %	1
4.	R1_S2	Generic	Ideal	Res= 820.0ohm Tolerance= 1%	1
5.	R2_S2	Generic	Ideal	Res= 1600.0ohm Tolerance= 1%	1

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